3.2 projective subvarieties of small codimension

3.2A Barth's Thm. $X \subseteq \mathbb{P}^r$ sm. irr proj subvariety of $\dim n$, $\operatorname{codim} e = r-n$

(i) $H^i(\mathbb{P}^r, \mathbb{C}) \to H^i(X, \mathbb{C})$ are isos for $i \le 2n-r = n-e$

(ii) more generally, $\pi_i(\mathbb{P}^r, X) = 0$ for $i \le 2n-r+1 = n-e+1$

$\Rightarrow H^i(\mathbb{P}^r, \mathbb{Z}) \to H^i(X, \mathbb{Z})$, $H_i(X, \mathbb{Z}) \to H_i(\mathbb{P}^r, \mathbb{Z})$ are iso. for $i \le 2n-r$

(Recall: ① $\pi_q(A) \to \pi_q(X) \to \pi_q(X, A) \to \pi_{q-1}(A) \to \cdots \to \pi_1(X, A) \to \pi_0(A)$)

① If (X, A) is k connected, A simply connected, non zero $\pi_j(X, A) = 0$ $1 \le j \le k$

② $H_i(X, A) = 0$ for $i \le k$ $\Leftrightarrow \pi_{k+1}(X, A) \simeq H_{k+1}(X, A)$

proof of 3.2.1 (i) ~~$j: X \to \mathbb{P}^r$~~, $j: X \hookrightarrow \mathbb{P}^r$. let $\xi \in H^2(\mathbb{P}, \mathbb{Z})$ the hyperplane class

$\bar{\xi} = j^*\xi \in H^2(X, \mathbb{Z})$, assume X has degree d. then the fundamental class of $\eta_X \in H^{2e}(\mathbb{P}^r)$

$H^*(\mathbb{P}^r, \mathbb{C}) = \mathbb{C}[\xi]/(\xi^{r+1})$ $\deg \xi = 2 \Rightarrow H^{2e}(\mathbb{P}^r, \mathbb{C}) = C \cdot \xi^e \Rightarrow \eta_X = a\xi^e$

$\int_{\mathbb{P}^r} \xi^r = 1$ $\int_{\mathbb{P}^r} \eta_X \cup \xi^n = a \int_{\mathbb{P}^r} \xi^r = a = \int_X \bar{\xi}^n$ (即 ξ 在 X 上限制 $\bar{\xi}$ 的自交数, 对应 d)

$= d$

$e = \operatorname{codim}_{\mathbb{P}^r} X$ $n = \dim X$

Consider the 'Gysin map' $j_*: H^i(X, \mathbb{C}) \to H^{i+2e}(\mathbb{P}^r, \mathbb{C})$

~~defined~~ defined by $H^i(X) \xrightarrow{PD} H_{2n-i}(X) \xrightarrow{j_*} H_{2n-i}(\mathbb{P}^r) \xrightarrow{PD^{-1}} H^{i+2e}(\mathbb{P}^r)$

i.e. $H^i(X) \xrightarrow{j_*} H^{i+2e}(\mathbb{P}^r)$

$\cong$ $H^{2n-i}(X)^* \xrightarrow{(j^*)^\vee} H^{2n-i}(\mathbb{P}^r)$ $\cong$

~~$\forall y$~~ $j^*j_*(y) = y \cdot j^*(\eta_X) = y \cdot d\xi^e$

$j_*j^*(x) = x \cdot \eta_X = x(d\xi^e)$ ②

take a tubular nbhd U of $X \subseteq \mathbb{P}^r$. U can be seen as a nbhd of zero section of $N_{X/\mathbb{P}^r}$.

$p: U \to X$ $p \circ j = id_X$ $j: X \to N_{X/\mathbb{P}^r}$ the zero section of $N_{X/\mathbb{P}^r}$. take a closed

Thom form $\tau \in \Omega^{2e}_c(U)$, with $\int_{p^{-1}(x)} \tau|_{p^{-1}(x)} = 1$ ($\forall x \in X$) (Compatible with the orientation of the complex Normal check)

(actually $\operatorname{Supp}\tau$ is proper over X) ~~we choose~~ τ is

take w a closed form representing y

$X \subseteq \mathbb{P}^r$ The hard part is $U \cong V \subseteq^{\text{open}} \mathbb{P}^r$

s.t. $V \cong X \times \mathbb{R}^{2e}$ s.t. $V \subseteq X \times \mathbb{R}^{2e}$ & $x \in V \rightsquigarrow (x, 0)$

$\downarrow$ is compatible

X

then we have $j_* \gamma = \gamma$. γ is extension by zero of $p^* w \wedge \tau \ (U \to P)$

Notice that $j_* \gamma$ is characterized by $\int_P j_*(\gamma) \cup b = \int_X \gamma \cup j^* b$

for all $b \in H^{2n-i}(\mathbb{P}^r, \mathbb{C})$ to prove $j_* \gamma = \gamma$. take β closed form representing b

$[\beta|_U] = p^* j^* [\beta] \Rightarrow \int_P (\gamma) \wedge \beta) = \int_U p^* w \wedge \tau \wedge \beta|_U = \int_U p^* (w \wedge j^* \beta) \wedge \tau$

$= \int_X w \wedge j^* \beta$ here uses $\int_{p^{-1}(x)} \tau|_{p^{-1}(x)} = 1$ locally this is Fubini's thm.

For the other. we use the language of cap product to prove.

$j: X \hookrightarrow \mathbb{P}^r$. $x \in H^i(\mathbb{P}^r)$, $\eta_x \in H^{2e}(\mathbb{P}^r)$ $j_*^{\text{hom}}(j^* x \cap c) = x \cap j_*^{\text{hom}} c \quad c \in H_n(X)$

$PD_{\mathbb{P}^r}(j_* j^* x) = j_*^{\text{hom}}(j^* x \cap [X])$ $j_*^{\text{hom}}[X] = \eta_X \cap [\mathbb{P}]$ this is how η_X is defined!

$\Rightarrow PD_{\mathbb{P}^r}(j_* j^* x) = x \cap (\eta_X \cap [\mathbb{P}]) = (\eta_X \cup x) \cap [\mathbb{P}] = (x \cup \eta_X) \cap [\mathbb{P}] = PD_{\mathbb{P}^r}(x \cup \eta_X)$

standard!

$\Rightarrow j_* j^* x = x \cup \eta_X$

$H^i(\mathbb{P}^r,\mathbb{C}) \xrightarrow{j^*} H^i(X,\mathbb{C})$

$\cdot d\xi^e \downarrow \qquad \nearrow i_* \qquad \downarrow \cdot d\bar{\xi}^e$

$H^{i+2e}(\mathbb{P}^r,\mathbb{C}) \xrightarrow[j^*]{} H^{i+2e}(X,\mathbb{C})$

if $n \le i \le ne$

by HL (3.1.40) $\cdot d\bar{\xi}^e : H^i(X) \to H^{i+2e}(X)$

is injective $\Rightarrow j_*$ is injective

(2)

$\cdot d\xi^e : H^i(\mathbb{P}^n,\mathbb{C}) \to H^{i+2e}(\mathbb{P}^r,\mathbb{C})$ is an isomorphism $\Rightarrow j_*$ is surjective

$\Rightarrow j_*$ is an iso $\Rightarrow j^* : H^i(\mathbb{P}^r) \to H^i(X)$ is an iso.

Rmk: Suppose $V \subset \mathbb{P}^N$ is sm complete intersection of dim r, $X \subseteq V$ sm subvar of dim n

由§3.1 易知 $H^i(\mathbb{P}^N) \to H^i(V)$ $\begin{cases} \text{iso} & i \le r \\ \text{inj} & i = r \end{cases}$ $e = r-n$

§3.2.B Hartshorne's Conj.

Let $X \subseteq \mathbb{P}^r$ be a sm. irr var of dim n If $3n > 2r \Rightarrow X$ is a complete intersection

($n >$ 2codim)

Rmk: $W = G(2,5) \to \mathbb{P}(\wedge^2 \mathbb{C}^5) = \mathbb{P}^9$ codim $= 3$ dim $= 6$ $(W \subseteq \mathbb{P}^9)$

W 在 $\mathbb{P}^9$ 中定义方程为5个，用矩阵写为 $\begin{pmatrix} u_0 & u_1 & u_2 & u_3 & u_4 \\ v_0 & v_1 & v_2 & v_3 & v_4 \end{pmatrix}$, $p_{ij} = u_i v_j - u_j v_i$

$0 \le i < j \le 4$

$G(2,5) \to \mathbb{P}^9$ $W \mapsto [p_{01} : p_{02} : \cdots : p_{34}]$

$p_{ij}p_{kl} - p_{ik}p_{jl} + p_{il}p_{jk}$ 这5个方程

e.g 3.2.10 $X \subseteq \mathbb{P}^r$ $\rho_k : H^0(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}(k)) \to H^0(X, \mathcal{O}_X(k))$ ρ_1 surjective

$=:$ X linearly normal. Conj: $X \subseteq \mathbb{P}^r$ sm subvar. of dim n if $3n > 2(r-1)$

$\Rightarrow$ linearly nml

e.g.: 在$\mathbb{P}^5$上找一个rank≥3的向量丛 s.t. $H^1(\mathbb{P}^5, E) \neq 0$ i.e. $E = \Omega^1_{\mathbb{P}^5}$ $H^1(\mathbb{P}^5, \Omega^1_{\mathbb{P}^5}) = H^{1,2}(\mathbb{P}^5) \neq 0$

choose $k \gg 0$ s.t. $k + c_1(E) > 0$ (i.e. $c_1(E) = c_1(\det E) = c_1(\mathcal{O}_{\mathbb{P}^5}(l)) = 1$ 即 Hyperplane) here $c_1(E)$ 为 Fulton 记号

Consider chose $e-1$ general sections $s_1, \dots s_{e-1} \in H^0(\mathbb{P}^5, E(k))$ $X = \{s_1 \wedge \cdots \wedge s_{e-1} = 0\}$ i.e. where $s_1 \dots s_{e-1}$ are linearly dependent i.e. $\varphi: \mathcal{O}_{\mathbb{P}^5}^{\oplus e-1} \to E(k)$ $(t_1 \dots t_{e-1}) \mapsto \sum_{i=1}^{e-1} t_i s_i$ is not surjective. locally φ is given by $e \times (e-1)$ matrix & X is defined by the vanishing of all of $(e-1)\times(e-1)$ minors $\Rightarrow X$ is closed subvariety of $\mathbb{P}^5$

Thm: Let $\mathcal{E}, \mathcal{F}$ be vector bundles on of rk m, n on sm. variety X / char 0 & $\mathcal{E}^\vee \otimes \mathcal{F}$ generated by global sections. If $\sigma: \mathcal{E} \to \mathcal{F}$ is a general morphism, then one of the following is true;

(1) $D_k(\sigma)$ is empty $\quad D_k(\sigma) = \{x \in X \mid \operatorname{rank}(\sigma(x)) \leq k\}$ (3)

(2) $D_k(\sigma)$ has expected codimension $(m-k)(n-k)$ & the sing locus is $D_{k+1}(\sigma)$

(Here general means $\exists$ Zariski open subset st. of $H^0(\mathcal{E}^\vee \otimes \mathcal{F})$ s.t. either all (1) or all (2))

Rmk 5: (1) $M_{m\times n}(\mathbb{C}) = \mathbb{A}^{mn}$ 考虑 consider $D_q = \{A \in M_{m\times n}(\mathbb{C}) \mid \operatorname{rank} A \leq q\}$ $0 \leq q \leq \min\{m,n\}$

D_q closed subvariety of $M_{m\times n}(\mathbb{C})$ $\operatorname{Codim} D_q = (m-q)(n-q)$ 考虑等于号 $\begin{pmatrix} B & C \\ D & F \end{pmatrix}$ $B \in M_{q\times q}(\mathbb{C})$

$\det B \neq 0 \rightsquigarrow \begin{pmatrix} B & 0 \\ 0 & F - DB^{-1}C \end{pmatrix}$ B, C, D 自由 $F = DB^{-1}C \Rightarrow \operatorname{Codim} D_q = (m-q)(n-q)$

(等于号成立 $\in$ 一个开集 $\Rightarrow$ $\operatorname{Hom}(\mathcal{E}, \mathcal{F})$ $\dim S \geq mn$

(2) Take $S = H^0(\mathcal{E}^\vee \otimes \mathcal{F})$ $S \otimes \mathcal{O}_X \twoheadrightarrow \mathcal{E}^\vee \otimes \mathcal{F}$ consider the total space morphism

$S \times X \twoheadrightarrow \operatorname{Tot}(\mathcal{E}^\vee \otimes \mathcal{F})$, take a local trivialization this becomes $U \times S \to U \times M_{m\times n}(\mathbb{C})$ $(p, \varphi) \mapsto (p, \varphi(p))$

Consider $T_q = \{(p, \varphi) \in X \times S, \operatorname{rank}\varphi(p) = q\}$

then locally $T_q = U \times D_q \times \mathbb{A}^{\dim S - mn}$ consider $p_0 \in U$ s.t. $\sigma_1, \dots \sigma_{mn} \in S$ restricts to a basis of $\operatorname{Hom}_{\mathbb{C}}(\mathcal{E}_{(p)}, \mathcal{F}_{(p)}) \Rightarrow$ ④ after possibly shrinking U $\sigma_1(p) \dots \sigma_{mn}(p)$ ~~is~~ gives a basis of $\operatorname{Hom}_{\mathbb{C}}(\mathcal{E}_{(p)}, \mathcal{F}_{(p)})$

~~Then~~ ~~we~~ i.e. if we consider $\sigma_i(p)$ as column vectors in $\mathbb{A}^{mn}$ $(\sigma_1(p), \cdots \sigma_{mn}(p))$ ~~gives~~ is non degenerate at every p, expand this to a basis of S $\sigma_1 \cdots \sigma_{mn}, \sigma_{mn+1} \cdots \sigma_N$

$N = \dim S$. ~~then we have~~ let $P(p) = (\sigma_{mn+1}(p) \cdots \sigma_N(s))$

then $\varphi_{(p)} = \sum_{j=1}^{mn} u_j \sigma_j + \sum_{i=1}^{N-ab} v_j \sigma_{mn+j}$ (u, v)

$(u_1 \cdots u_{mn} v_1 \cdots v_{N-mn})$ is the coordinate of φ under the base $\{\sigma_i\}$

$u = (u_1 \cdots u_{mn}) \in \mathbb{A}^{mn}$ $v = (v_1 \cdots v_{N-mn}) \in \mathbb{A}^{N-mn}$ u.v. 表示对应的坐标

we have a local isomorphism $U \times S \longrightarrow U \times M_{mn}(\mathbb{C}) \times \mathbb{A}^{N-mn}$

$(p, (u, v)) \longmapsto (p, Q(p)u + R(p)v, v)$ gives an iso.

since $(p, w, v) \in U \times M_{mn}(\mathbb{C}) \times \mathbb{A}^{N-mn} \longmapsto (p, Q(p)^{-1}(w - R(p)v), v)$ is an inverse

under this iso the projection only restrict the ~~w~~ ~~to~~ second coordinate.

$\Rightarrow \dim T_{q_h} = \dim X + N - (m-q_h)(n-q_h)$ $T_{q_h} \xrightarrow{p} S$ the fiber over $\varphi \in S$ is exactly

$\{p \in X \mid \operatorname{rank} \varphi(p) = q_h\}$ by generic smooth applied to dominant component over of T_q / S

(if $\exists$) $\Rightarrow \exists U \subset S$ s.t. fiber over U is smooth. $\dim T_{q_h} - \dim S = \dim M - (m-q_h)(n-q_h)$

~~See~~ For Connectedness & Nonemptiness of a fixed $u \in \operatorname{Hom}(E, F)$, see II. Thm 2.1

(4)

The Eagon-Northcott complexes on X

$\varphi: E \to F$ be a morphism of vector bundles of rank $m \geq n$ ~~then~~ if X is smooth & if $Z = D_{n-1}(\sigma)$ has expected codim $m-n+1$ $\Rightarrow EN_k$ is acyclic

write down EN. we have $0 \to \mathcal{O}_{\mathbb{P}^r}^{e-1} \longrightarrow E(k) \to I_X(a) \to 0$ $a = ek + C_1(E)$

~~where~~ (we can transpose the $E \to \mathcal{O}$ to $\mathcal{O}^\vee \to E^\vee$) $I_X \subset \mathcal{O}_{\mathbb{P}^r}$ the

$0 \to H^0(\mathbb{P}^5, \mathcal{O}_{\mathbb{P}^5}(4-k)) \to H^0(X, \mathcal{O}_X(4-k)) \to H^1(I_X(4-k)) \to 0$

It's linearly normal by Zak's thm by not projectively normal!

$\Rightarrow H^1(\mathcal{O}_{\mathbb{P}^5}(4-k))$

Now consider $X \subseteq \mathbb{P}^{n+2}$ sm closed subvar of dim n by 3.2.3, if $n \geq 4$

take $\Rightarrow \omega_X = \mathcal{O}_X(k)$ for some $k \in \mathbb{Z}$, by Serre's construction. $\exists$ vector bundle of rk 2 E

$s \in H^0(E)$ s.t. $X = \text{Zeroes}(s)$ & X is complete intersection $\Leftrightarrow E$ splits

Then If Conj 3.2.8 holds for codim 2 $\Rightarrow$ $\mathbb{P}^k$ on $\mathbb{P}^r$, very vector bundle E of rk 2 on $\mathbb{P}^r$ splits

Hartshorne's splitting conj.

$\Leftarrow$ ✓

Rmk 3.2.12

Evans & Griffith: E a vector bundle of rk e on $\mathbb{P}^r$ having property that

$H^i(\mathbb{P}^r, E(k)) = 0$, for all $k \in \mathbb{Z}$. then $\exists$ $e-1$ sections $\Rightarrow E$ splits as a sum of line bundles

by this thm, we can show $X \subseteq \mathbb{P}^{n+1}$ a sm closed subvar of dim n is a complete intersection

$\Leftrightarrow X$ is projectively normal (if $n \geq 4$)

complete intersection $\Rightarrow$ suppose X is the complete intersection of hypersurfaces of degree d_1 & d_2

& X can be seen as the zero section of (f_1, f_2) $H^0(\mathbb{P}^{n+2}, \mathcal{O}(d_1) \oplus \mathcal{O}(d_2))$ $\deg f_1 = d_1$ $\deg f_2 = d_2$ writing its Koszul

complex we have $0 \to \mathcal{O}_{\mathbb{P}^{n+2}}(-d_1-d_2) \to \mathcal{O}_{\mathbb{P}^{n+2}}(-d_1) \oplus \mathcal{O}_{\mathbb{P}^{n+2}}(-d_2) \to I_X \to 0$ twist by $\mathcal{O}(m)$

$\Rightarrow H^1(\mathbb{P}, I_X(m)) = 0$ $\Rightarrow$ projectively normal; by 3.2.3, $\omega_X \cong \mathcal{O}_X(k)$. by Serre's construction we have a

rank 2 E on $\mathbb{P}^{n+2}$, $X = \text{Zeroes}(s)$ $s \in H^0(E)$. by Koszul or EN$_0$ we have $0 \to \mathcal{O}_{\mathbb{P}^{n+2}} \to E \to I_X(a) \to 0$

$(\det E) = \mathcal{O}_{\mathbb{P}^{n+2}}(a)$ (1. $E|_X = N_{X/\mathbb{P}^{n+2}}$ 2. $\mathcal{O}_X \omega_X = \omega_{\mathbb{P}^{n+2}}|_X \otimes \det N_{X/\mathbb{P}^{n+2}}$) $a = k+n+3$

$E \cong \det E \otimes E^\vee$

projectively normal $\Rightarrow H^1(\mathbb{P}^{n+2}, I_X(m)) = 0$ (since $H^1(\mathbb{P}^{n+2}, \mathcal{O}(m)) = 0$ $\forall m \in \mathbb{Z}$)

($m \geq 1$ by projectively normal $m \leq 0$ 'automatic')

then by $0 \to \mathcal{O}_{\mathbb{P}^{r-d+1}} \to \mathcal{E} \to \mathcal{I}_X(e) \to 0 \Rightarrow H^i(\mathbb{P}^r, \mathcal{E}(k)) = 0 \ \forall k \in \mathbb{Z} \Rightarrow \mathcal{E}$ 的 $\Rightarrow$

Complete intersection

Cor 3.2.2 : $X \subseteq \mathbb{P}^r$ sm variety of dim n. If $2n - r \geq 1 \Rightarrow X$ simply connected

$\Rightarrow \pi_2(\mathbb{P}^r, X) \to \pi_1(X) \to \pi_1(\mathbb{P}^r) \to \cdots$ 事 $\mathbb{P}^2$ 中相交的光滑曲线 X elliptic curves $\pi_1(X) = \mathbb{Z}^2$

Cor 3.2.3 : $X \subseteq \mathbb{P}^r$, sm sub of dim n If $2n+3 \geq 2$ then 即 $\mathrm{Pic}(\mathbb{P}^r) \to \mathrm{Pic}(X)$

gives an iso. $H^i(\mathbb{P}^r, \mathbb{Z}) \to H^i(X, \mathbb{Z})$ are isos for $i \leq 2$ by thm 3.2.1

It results from Hodge theory $H^i(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}) \to H^i(X, \mathcal{O}_X)$ ~~then~~ are isos $i \leq 2$

由 $0 \to \mathbb{Z} \to \mathcal{O}_X \to \mathcal{O}_X^* \to 0$ ~~$H^0(\mathcal{O}_X)$~~ $\to H^1(X, \mathcal{O}_X) \to \mathrm{Pic}(X) \to H^2(X, \mathbb{Z}) \to H^2(X, \mathcal{O}_X)$

$\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$

$H^2(\mathbb{P}^r, \mathbb{Z}) \to H^1(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r}) \to \mathrm{Pic}(\mathbb{P}^r) \to H^2(\mathbb{P}^r, \mathbb{Z}) \to H^2(\mathbb{P}^r, \mathcal{O}_{\mathbb{P}^r})$

by I- Lemma ~~counter example~~ This is also sharp : $\mathbb{P}^3$ 中光滑 2 次曲面 $\mathbb{P}^1 \times \mathbb{P}^1 \subseteq \mathbb{P}^3$

$\mathbb{Z} \oplus \mathbb{Z} \quad \mathbb{Z}$.

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§3.3 Connectedness thms

§3.3 A. Bertini's thms

Thm 3.3.1 : X irr. var. $f : X \to \mathbb{P}^r$ morphism. fix $d < \dim f(X)$ if $L \subseteq \mathbb{P}^r$ is a general

$(r-d)$ plane then $f^{-1}(L)$ is irr.

(在 $r-d+1$ 维线性空间之间的 of $\mathbb{C}^{r+1}$)

Pf:

Lemma : $p : V \to T$ dominating morphism of irr. ~~quasi~~ cpx varieties if p admits a ~~smooth~~ rational section $s : T \to V$

($\forall p$ $p \circ s = \mathrm{id}_T$) whose image does not lie in the singular locus of V i.e.

$s(t) \in V - \mathrm{sing}(V)$ for general $t \in T$

then $U_t = p^{-1}(t)$ is irr for general $t \in T$

pf: By dimension of fibers If we take $Z \subseteq$ ~~V~~ $\coprod Z$ open $Z = Z_1 \cup \dots \cup Z_N$

By thm on

the dimension of general fibers of $f|_{Z_i} \to \overline{f(Z_i)}$ is $\leq r-1$ r = relative dimension $\dim V - \dim T$

$\Rightarrow$ general fiber of p are pure dimensional of dim r

But

$\Rightarrow$ for general $t \in T$, irr compts of U_t are in 1-1 correspondence with those of V_t

For this reason we may assume V, T are smooth & p is a smooth map

by Ehresmann's thm p is topologically locally trivial ($\forall t \in T$ $\exists$ $U_t \subseteq T$ open nbhd of t s.t. $p^{-1}(U_t) \cong p^{-1}(t) \times U_t$ s.t $p^{-1}(U_t) \overset{\Phi}{\cong} p^{-1}(t) \times U_t$

$V_{(t)} \downarrow \Phi \swarrow p_2$

U_t

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$\forall t \in T$. $\exists$ $p^{-1}(t)$ 中它含($\subseteq$)的 connected comp 记为 $Y_{(t)}$, $\exists$ $W = \bigcup_{t \in T} C_{(t)} \subset V$

~~存在 $p^{-1}(U_t)$ 上~~

s can be seen as $U_t \to p^{-1}(t)$

for a fixed t ($\in U_t$) take U_t to be connected, ~~$S(U_t)$~~ $\forall t' \in U_t$, $Y_{(t')} \overset{\Phi}{\leftrightarrow} C$

$S(U_t)$ is contained in a connected comp of $p^{-1}(t)$, then $A = Y_{(t)} \Rightarrow \Phi(W \cap p^{-1}(U_t))$

$= U_t \times A \Rightarrow W|_{p^{-1}(U_t)}$ is open & closed $\Rightarrow W = V \Rightarrow \forall t$ $Y_{(t)} = p^{-1}(t)$ $\Rightarrow$ connected.

proof of 3.3.1 : First assume f is generically finite. $n = \dim X = \dim \overline{f(X)}$, it suffices to prove the case $d = n-1$

proof of this :

lemma : Suppose T ~~finite~~ variety/$\mathbb{C}$, $h: T \to \mathbb{P}^N$ a morphism $0 \leq c \leq N$ then for general codim c linear subspace $M \subseteq \mathbb{P}^N$ (by these general means non empty open subset in the corresponding Grassmann)

定理 3.3.1：到边界情形的严格归约

命题. 设 X 是不可约复代数簇，$f\colon X \to \mathbf{P}^r$ 是在其像上 generically finite 的态射，记

$$n = \dim X = \dim \overline{f(X)} \geq 1.$$

假设已经知道：对一般的余维 $n-1$ 线性子空间 $L' \subseteq \mathbf{P}^r$，$f^{-1}(L')$ 非空且不可约。那么，对每个整数 $0 \leq d < n$，一般的余维 d 线性子空间 $L \subseteq \mathbf{P}^r$ 的逆像 $f^{-1}(L)$ 也非空且不可约。

这里的“一般”均指相应 Grassmannian 的某个非空 Zariski 开集上的所有参数。我们不假设 X 完备或光滑；逆像取概形论逆像，但不可约性及不可约分支均指其底层拓扑空间。

论证要点. 需要建立三个事实：一般的 $f^{-1}(L)$ 的每个不可约分支 C_i 都满足 $\dim C_i = \dim \overline{f(C_i)} = n-d$；一般的 $L' \subset L$ 的逆像会与 $C_i \setminus \bigcup_{j \neq i} C_j$ 相交；旗簇保证可以同时要求这样的 L' 属于边界假设中的好参数开集。最后才使用 $f^{-1}(L')$ 的不可约性产生矛盾。

一、一般线性条件的维数估计

引理. 设 T 是有限型复代数概形，$h\colon T \to \mathbf{P}^N$ 是态射，$0 \leq c \leq N$。则对一般的余维 c 线性子空间 $M \subseteq \mathbf{P}^N$，

$$\dim h^{-1}(M) \leq \dim T - c. \tag{1}$$

若右边为负，则结论解释为 $h^{-1}(M) = \varnothing$。此外，若 $T = Z \subseteq \mathbf{P}^N$ 是非空闭子集且 h 是包含映射，则在 $\dim Z \geq c$ 时，对一般的 M 有

$$Z \cap M \neq \varnothing, \qquad \dim(Z \cap M) = \dim Z - c. \tag{2}$$

证明. 先设 T 非空且不可约，并赋予它约化结构。记

$$G = \mathrm{Gr}(N-c+1, N+1), \qquad I_T = \{(t, M) \in T \times G : h(t) \in M\}.$$

$I_T \to T$ 是一个 Grassmannian 丛：若点 $h(t)$ 对应于直线 $\ell_{h(t)} \subset \mathbf{C}^{N+1}$，则包含该点的 M 对应于 $\mathbf{C}^{N+1}/\ell_{h(t)}$ 中的 $N-c$ 维子空间。因此纤维为 $\mathrm{Gr}(N-c, N)$。这一描述来自 h 拉回的秩 N 商向量丛，故该丛在 Zariski 拓扑下局部平凡。于是 I_T 不可约，且

$$\dim G = (N-c+1)c,$$

$$\dim I_T = \dim T + (N-c)c = \dim T + \dim G - c.$$

考虑第二投影 $I_T \to G$。若它不占优，去掉其像的闭包后，所有纤维为空。若它占优，由纤维维数定理，在 G 的某个非空开集上有

$$\dim h^{-1}(M) = \dim I_T - \dim G = \dim T - c.$$

这证明了不可约情形。一般情形对 T 的有限多个不可约分支分别应用上述结论，再取相应稠密开集的交，即得 (1)。此处不要求 T 完备。

若 $Z \subseteq \mathbf{P}^N$ 是闭子集且 $\dim Z \geq c$，则射影维数定理给出：对每个余维 c 的 M，

$$Z \cap M \neq \varnothing, \qquad \dim(Z \cap M) \geq \dim Z + \dim M - N = \dim Z - c.$$

该下界亦可由仿射锥证明：Z 的仿射锥维数为 $\dim Z + 1$，施加定义 M 的 c 个齐次线性方程后，Krull 高度定理给出维数至少为 $\dim Z + 1 - c \geq 1$；所得锥不只包含原点，射影化后即得上述结论。与 (1) 合并，得到 (2)。 □

二、边界情形推出一般情形

命题的证明. $d=0$ 时 $f^{-1}(\mathbf{P}^r)=X$，结论成立；$d=n-1$ 正是所作假设。因此以下固定 $1\le d\le n-2$。若没有这样的整数，则无需再证明。

第一步：用旗簇明确两个“一般”的量词。 对 $e=d,n-1$，记 $G_e=\mathrm{Gr}(r-e+1,r+1)$。边界假设给出非空开集 $\Omega\subseteq G_{n-1}$，使每个 $L'\in\Omega$ 均满足 $f^{-1}(L')$ 非空且不可约。考虑旗簇

$$F=\{(L',L)\in G_{n-1}\times G_d : L'\subset L\}$$

记两个投影为 $p\colon F\to G_d$，$(L',L)\mapsto L$，以及 $q\colon F\to G_{n-1}$，$(L',L)\mapsto L'$。两个投影都是 Grassmannian 丛，因而都是光滑满射；特别地，p 是开映射。故

$$V=p\big(q^{-1}(\Omega)\big)\subseteq G_d$$

是非空开集，因 G_d 不可约而稠密。对每个 $L\in V$，

$$p^{-1}(L)\cap q^{-1}(\Omega)\quad \text{是}p^{-1}(L)\text{ 的非空开子集.} \tag{3}$$

而 $p^{-1}(L)$ 是一个不可约 Grassmannian，参数化 L 中所有余维 $n-1-d$ 的线性子空间 L'。因此 (3) 也是该纤维中的稠密开集。此外，每个 $L\in V$ 都包含某个 $L'\in\Omega$，所以 $f^{-1}(L)$ 非空。

第二步：控制 $f^{-1}(L)$ 的每个分支及其像的维数。 由于 f 在其像上 generically finite，存在非空开集 $X^\circ\subseteq X$，使 $f|_{X^\circ}$ 为 quasi-finite。令 $B=X\setminus X^\circ$，则 $\dim B\le n-1$。将引理分别用于 $f\colon X\to\mathbf{P}^r$ 和 $f|_B$。将 V 与所得的两个稠密开集相交后，仍记为 V；它仍是非空开集，且对每个 $L\in V$ 有

$$\begin{aligned}\dim f^{-1}(L)&\le n-d,\\ \dim\big(B\cap f^{-1}(L)\big)&\le n-d-1.\end{aligned} \tag{4}$$

当 $B=\varnothing$ 时，第二个条件自动成立。

固定任意 $L\in V$，写出其逆像的不可约分支分解

$$f^{-1}(L)=C_1\cup\cdots\cup C_s.$$

L 由 d 个齐次线性方程定义；在 $f^*\mathcal{O}_{\mathbf{P}^r}(1)$ 的局部平凡化中，$f^{-1}(L)$ 由 d 个正则函数定义。由 Krull 高度定理，每个 C_i 都满足 $\dim C_i\ge n-d$。结合 (4)，得到

$$\dim C_i=n-d,\qquad C_i\not\subseteq B. \tag{5}$$

因此 $C_i\cap X^\circ$ 是 C_i 的非空稠密开子集。$f|_{C_i\cap X^\circ}$ 为 quasi-finite，且稠密开子集的像在整个像中稠密。故由纤维维数定理，

$$\dim\overline{f(C_i)}=\dim\overline{f(C_i\cap X^\circ)}=\dim(C_i\cap X^\circ)=n-d. \tag{6}$$

这里以及下文的像闭包均在 L 中取。

8.3

第三步：使 $f^{-1}(L')$ 遇到每个分支独有的部分。对每个 i，令

$$U_i = C_i \setminus \bigcup_{j\neq i} C_j, \qquad Z_i = \overline{f(C_i)} \subseteq L.$$

由于 C_i 是不可约分支，U_i 是 C_i 的非空稠密开子集。因此 $f(U_i)$ 是 Z_i 的稠密构造性子集：稠密性来自 U_i 在 C_i 中稠密，构造性来自 Chevalley 定理。故存在真闭子集 $F_i \subsetneq Z_i$，使

$$Z_i \setminus F_i \subseteq f(U_i), \qquad \dim Z_i = n-d, \qquad \dim F_i \leq n-d-1. \tag{7}$$

允许 $F_i = \varnothing$。

在 $L \cong \mathbf{P}^{r-d}$ 中，将引理用于 Z_i 和 F_i，并取 $c = n-1-d$。于是对 $p^{-1}(L)$ 中的一个稠密开集上的每个 L'，

$$\begin{aligned} &\dim(Z_i \cap L') = (n-d)-(n-1-d) = 1, \\ &\dim(F_i \cap L') \leq (n-d-1)-(n-1-d) = 0. \end{aligned} \tag{8}$$

第二行允许交集为空。由此 $Z_i \cap L'$ 不可能包含于 F_i，所以

$$(Z_i \setminus F_i) \cap L' \neq \varnothing.$$

结合 (7)，可取 $z \in f(U_i) \cap L'$，再取 $x \in U_i$ 满足 $f(x) = z$。于是

$$\left(C_i \setminus \bigcup_{j\neq i} C_j\right) \cap f^{-1}(L') = U_i \cap f^{-1}(L') \neq \varnothing. \tag{9}$$

对有限多个 i 取相应稠密开集的交，再与 (3) 中的稠密开集相交。因 $p^{-1}(L)$ 不可约，该交仍非空。因此可以选择同一个 $L' \subset L$，使 (9) 对所有 i 成立，且 $L' \in \Omega$。

第四步：利用边界假设排除多个不可约分支。对刚才选定的 L'，边界假设保证 $f^{-1}(L')$ 非空且不可约。由于 $L' \subset L$，

$$f^{-1}(L') = \bigcup_{i=1}^{s} \left(C_i \cap f^{-1}(L')\right).$$

右侧是有限个闭子集的并。因此存在 i_0 使

$$f^{-1}(L') \subseteq C_{i_0}.$$

如果 $s \geq 2$，取 $j \neq i_0$。由 (9)，存在

$$x \in \left(C_j \setminus \bigcup_{i\neq j} C_i\right) \cap f^{-1}(L').$$

一方面 $x \notin C_{i_0}$，另一方面 $x \in f^{-1}(L') \subseteq C_{i_0}$，矛盾。故 $s = 1$，即 $f^{-1}(L)$ 不可约。

上述论证适用于同一个非空开集 $V \subseteq G_d$ 中的每个 L，故证明了对一般余维 d 的线性子空间的结论。这就完成了从 $d = n-1$ 的边界情形到所有 $0 \leq d < n$ 的归约。 □

关于假设的作用. Generically finite 假设用于第二步，使每个一般逆像分支的像与该分支具有相同维数。对于非完备的 X，$f(C_i)$ 不必是闭集，因此第三步必须使用 $Z_i = \overline{f(C_i)}$ 以及 $Z_i \setminus F_i \subseteq f(U_i)$，不能直接把 $f(C_i)$ 当作射影闭子簇。

出处. 本命题补全 R. Lazarsfeld，*Positivity in Algebraic Geometry I*，Theorem 3.3.1 的证明开头所述归约（印刷页 207）。文中只将边界情形作为假设，不使用一般余维情形的 Bertini 不可约性定理。

Theorem 3.3. X irr. $f: X \to \mathbb{P}^r$ morphism, $L \subseteq \mathbb{P}^r$ an arbitrary linear sp. of codim $d < \dim \overline{f(X)}$. If X is complete $\Rightarrow f^{-1}(L)$ is connected

If X not complete but f is proper over $V \subseteq \mathbb{P}^r$ (zariski open set) s.t. $L \subseteq V$

Pf: let $B = \mathbb{P}^r \setminus V$ $I_B = \{(y, L') \in B \times G : y \in L'\}$ $G = G(r-d+1, r+1)$

I_B closed in $B \times G$. $\pi_2 : I_B \to G$ proper $\Rightarrow \pi_2(I_B)$ closed $W = G \setminus \pi_2(I_B)$

($\pi_2(I_B)$ 是那些与 B 相交的那些 $M \in G$, 因为 $B \cap M$ 中点满足 $\in M$)

$\Rightarrow W$ open. $Z := \{(x, L') \mid f(x) \in L'\} \subseteq X \times W \subseteq X \times G$, as before it's a Grassmannian

bundle over $X \Rightarrow Z$ irreducible $Z \to W$ is obtained from $\tilde{Z} \to G$ (restriction of $\hat{f}$)

$\tilde{Z} = \{(x, L') \in X \times G \mid f(x) \in L'\} \Rightarrow Z \to W$ proper!

take Stein factorization: $Z \xrightarrow{q} W' \xrightarrow{r} W$ q has connected fibers & r finite

by 3.3, general fiber of p_2 is irreducible $\Rightarrow r$ is generically one to one

but r is $\Rightarrow r$ is birational on W finite $\Rightarrow r$ is isomorphism $\square$.

e.g. 3.3.5. X irr. quasi-projective. $\forall x_1, x_2 \in X$ distinct, $\exists$ irr curve $\Gamma \subseteq X$ s.t. $x_1, x_2 \in \Gamma$

& $\exists$ map $: T \to X$ T sm. curve whose image contain x_1, x_2

take $X' = Bl_{x_1, x_2}(X)$, E_1, E_2 exceptional divisors

embed X' in a large $\mathbb{P}^N$ st. $\exists x \in \mathbb{P}^N$ ~~$\overline{\bigcup_{t \in [E_1]} t}$~~ st. $x \cap \overline{X'} \cap \mathbb{P}^N$ is empty

then ~~take projection from x restrict to X' gives $f: X' \to \mathbb{P}^r$ generically finite, finite~~ along E_i

then quasi

~~take a general~~ since X' is quasi-projective $\exists$ locally closed embedding $X' \hookrightarrow \mathbb{P}^n$

by 3.3, take a general linear sp of codim $\dim X - 1$ $\Gamma' = f^{-1}(L)$ is irreducible of dim 1